Evolving a Printable Titanium Karambit under Ergonomic, Edge and Serration Constraints

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***Abstract*—We report the best individual of the G3 v3.0.0 cohort of record, a one-piece Ti-6Al-4V karambit for laser powder-bed fusion with a hollow, lattice-filled grip and blade core. No person designed it: 24 independent CVT-MAP-Elites islands ran 2,500 evaluations each from scrambled Sobol samples, and the design space itself carries the rules a good edge must follow. Both cutting edges have one primary grind of 20.5–20.8° and 20.6–21.6° per side, measured normal to the edge and carried to within 3.0 and 3.1 mm of the point, and both edges are serrated (3 and 5 teeth; the weakest tooth has a safety factor of 2.83 under a 180 N snag). The design scores 80.55 under W_ERGOLIGHT_v1 draft d with an ergonomic multiplier of 0.988 and weighs 59.3 g as printed, which is above the 50 g goal. The 0.4 mm search grid undercounted the mass of the search's own leader (51.3 g against 75.2 g at 0.2 mm and 74.6 g at 0.15 mm), so the ranking of record is measured on the 0.2 mm grid of the print file. Against a safety-factor target of 1, its solid tip (0.64) is below the target and its grip under the spine load (0.74 by voxel FE) is below the target. We describe the representation, objective, search and measurement chain, the evolution that produced the design, and every limit of the result.**

***Index Terms*—Evolutionary design, quality diversity, MAP-Elites, additive manufacturing, Ti-6Al-4V, finite elements, knife design.**

1 Introduction

A karambit is a curved, ring-handled knife held in a closed fist with the little finger through the ring. Printing one in Ti-6Al-4V by laser powder-bed fusion makes a hollow, lattice-filled part possible, which opens a large design space but also makes it easy to fool oneself: a coarse voxel model can hide mass, a strength formula can credit material that is never built, and an edge can be left unground where a test does not look. G3 is the generation of the Jarambit project that designs the whole knife around the hand. This paper reports its v3.0.0 release.

Three choices define the release. First, *evolution designs the knife*: populations start from scrambled Sobol samples, nobody edits an individual, and engineering requirements enter as constraints on what the decoder can express or as measured terms of the objective. Second, *the edge is part of the design space, not a post-process*: both cutting edges must carry one primary grind at a knife-appropriate angle, run unbroken to the point, and be serrated with teeth that do not snap (Section 2). Third, *only measurements are reported*: every number in this paper and in the repository README is generated from one release receipt, and the strength figures quoted for the design come from voxel-SDF geometry, plate and solid finite elements, and a voxel finite-element certification of the printed geometry.

The contributions are the decoder rules for edges and serrations (EDGE-1..5, SERR-1..4), the logged evolution of a 24-island cohort, the design it found, and a measurement chain whose every step is reproducible from a commit and a genome row (Section 8).

Release v3.0.0, decoder commit `80e669e5b558` (decoder hash `a203278d80d2`). All figures and numbers in this paper are generated from the release receipt evidence/g3/v3.0.0/release.json. Status: computational design; not built, printed or physically tested.

2 Design space

2.1 Genome and decoder

A G3.0 genome is 179 real genes in the unit cube (g3_genome.rs). The decoder maps them to physical parameters and builds a signed distance field (SDF) of the whole knife (g3_sdf.rs): a finger ring with a fixed 24.9 mm bore and 8 mm lip, a compact grip with finger grooves and a textured surface, a guard, and a curved claw blade whose straight-line reach from the guard is capped at 85.725 mm. The interior of the grip and blade is a skin over a triply periodic minimal-surface lattice with powder-removal ports, so the part can be printed hollow. The decoder revision of this release, r2, is identified by the hash `a203278d80d2` of its source files; an archive made under another decoder fails its integrity check.

2.2 Edge rules (EDGE-1..5)

Each cutting edge has one primary grind whose per-side angle is a gene bounded to 17.5–22.5° (35–45° inclusive, centred on 20° as a hard-use edge for titanium). The bevel depth is the true distance to the actual, serrated edge polyline, so the angle is the same on the curved claw as on the straight part, and the grind is carried at full height until it meets the blade body, with no micro-bevel and no shoulder (EDGE-1, -2, -3). Both grinds run unbroken to the point; only a plunge of 5–12 % of the edge length next to the guard is unsharpened, inside a 4.0 in sharpened-length cap per edge (EDGE-4). The sharpness check reads the true edge-normal section over the first 2 mm at every steady station, so a shoulder behind the apex would cost score (EDGE-5).

2.3 Serration rules (SERR-1..4)

Both edges are serrated. The serration depth is 0.8–2.2 mm on the inner edge and 0.8–2.0 mm on the spine edge (SERR-1); every serrated span holds at least three full-depth teeth (SERR-2); the span starts past the plunge and ends at least 18 mm from the point, so the piercing point stays plain (SERR-3). To keep teeth from snapping, the pitch is 5–9 mm and the depth is at most a quarter of the pitch (SERR-4). Each tooth is checked as built (serration_tooth_strength): its root is the segment between two neighbouring gullet bottoms, with the

thickness of the exact blade law along it; a 180 N snag along the edge, amplified by the Inglis notch factor $K_t = 1 + 2\sqrt{d/\rho}$ for gullet depth d and radius ρ [6], and a 60 N pry across the blade load the tooth tip. The decoder test suite requires every tooth of 60 Sobol designs to reach a safety factor of at least 1 under both cases.

3 Objective

Designs are ranked by W_ERGOLIGHT_v1, draft d (ergolight.rs):

$$\text{score} = 100\,E \sum_i w_i\, s_i, \qquad E = \prod_j e_j^{\beta_j}, \tag{1}$$

where $s_i \in [0, 1]$ are term scores with weights w_i summing to 1, and E is a geometric-mean ergonomic multiplier over eight hand-zone factors e_j (ring lip, grip section, hand-edge radii, finger index, grip length, guard, grip surface and balance). Mass carries the largest weight, $w = 0.370$, so lighter is always better; the solid tip carries 0.070; the remaining terms cover lateral and ring capacities, blade and human-maximum twist, spine load, buckling, fatigue, failure sequence, sharpness and point profile, hollow-grip structure, watertightness, karambit form, fairness and buildability. No term is a hard gate: a design below a target loses score in proportion, and the search can still learn from it. Safety-factor terms use a saturating map that rewards factors up to about 1 and little beyond.

4 Search

Each island is a CVT-MAP-Elites archive [1], [2] of 1,024 niches over eight measured behaviour descriptors: hook drop, inner turn angle, silhouette complexity, grip roundness, groove articulation, porosity, surface energy and mass (g3_qd.rs). An island starts from 64 scrambled Joe–Kuo Sobol samples [3], [4]. Afterwards each child is bred from two archived parents by iso-line variation: 40 % of parent pairs come from the best-scoring elites, the rest from a curiosity-weighted tournament of three that favours niches whose offspring recently improved the archive. A child takes its niche if the niche is empty, if it was evaluated at a higher tier than the incumbent, or if it scores higher at the same tier.

Evaluation is a three-tier cascade. Tier 0 is analytic and takes milliseconds. A design scoring at least 62 goes to Tier 1: a voxel SDF of the whole knife at $h = 0.4$ mm, its mass and inertia, and plate finite elements of the blade and ring. A Tier-1 score of at least 71 promotes it to Tier 2, which adds a watertight mesh census and a solid TET10 model of the point. Only Tier-1 and Tier-2 results are measurements; Tier-0 values never appear in this paper.

The islands are independent, with distinct seeds, one thread each. Every island logs its progress every 50 evaluations: the best measured score, archive size, Tier-2 count, and at power-of-two evaluation counts the best genome itself, which is what Section 6 plots and renders. Logging draws no random numbers, so a logged run produces the same archive as an unlogged one.

5 Measurement and verification

Table 1 | Safety factors of the winner: search surrogates and voxel finite elements of the printed geometry

Load case	Grip formula	Grip, voxel FE	Root, voxel FE
Lateral 260 N	5.07	3.55	2.08
Twist 16.2 N·m	12.69	5.50	1.69
Spine 3427 N	0.90	0.74	1.29

5.1 Platform and grid

The cohort ran on xbabe4 (arm64 macOS). Floating-point libraries differ between platforms, so the final ranking was repeated on Linux x86_64 (xbabe2), Tier 2 at h 0.2 mm, which is the platform of record for every number below; the winner scored 80.55 on the search host and 80.55 on the platform of record. The search grid is $h = 0.4$ mm and the ranking grid $h = 0.2$ mm (Section 6). Mass is reported from the watertight 0.2 mm mesh of the printed part (13,385 mm^3 at 0.00443 g/mm^3, 59.3 g); the ranking counted 57.0 g on its voxel grid and the search 56.0 g.

5.2 Solid tip

The point is checked by a solid TET10 model of the last 30 mm of the blade under a 1.82 N·m pry, reading the peak nodal von Mises stress 4–15 mm from the point. Its planform mesh was refined until it converged on sharp points: for this design the safety factor is 0.585, 0.638 and 0.611 at maximum triangle areas of 0.4, 0.15 (production) and 0.08 mm^2.

5.3 Voxel finite-element certification

The plate and beam models of the search are fast surrogates. For the winner we also solved the printed geometry itself: a hex8 linear-elastic model on its AM-build voxels at $h = 0.4$ mm (skin, lattice, ribs and ports as built), in two zones, the grip and the blade root. Each zone is clamped at one end and loaded at the other with the wrench of three cases: 260 N lateral at the tip, the 16.2 N·m human-maximum twist, and 3427 N on the spine at the blade centroid. We report the 99.9th-percentile von Mises stress of each zone against an allowable of 674 MPa, not a single-element peak, because the voxel staircase concentrates stress artificially at re-entrant corners. Table 1 compares the two.

6 Results

6.1 The evolution

The cohort ran 24 islands of 2,500 evaluations each, 60,000 in all, from 2026-10-26T11:03:49Z to 2026-10-26T12:32:55Z UTC, and kept 3,177 elites, 68 of them at Tier 2. Fig. 1 shows the best measured score of every island against evaluations, and how the archives filled. The winner came from island island13, which started from a best measured score of 56.54 after its Sobol initialisation. Fig. 2 shows that island's best design at each logged checkpoint, from the same camera.

6.2 Choosing the winner on the print grid

The search measures every design on an $h = 0.4$ mm grid, and that grid can undercount mass. The search's own leader, `hz4i013_g00840_2a44149e` (Tier-2 score 81.53 at $h = 0.4$ mm), weighs 51.3 g at $h = 0.4$ mm but 75.2 g at $h = 0.2$ mm and

Table 2 | Grid check: mass (g) of the top 8 of the search-grid ranking at five voxel pitches h (mm), Tier 1

Design	0.4	0.3	0.25	0.2	0.15
`hz4i013_g00840_2a44149e`	51.3	55.6	54.9	75.2	74.6
`hz4i013_g00995_d206e3c9`	56.0	60.0	59.0	57.0	57.5
`hz4i002_g00995_6f4643f1`	58.8	59.6	58.9	58.7	58.5
`hz4i013_g00345_be45c4bc`	55.3	59.7	56.4	56.7	57.1
`hz4i002_g00479_76f5e0ac`	58.0	61.7	58.2	59.5	58.9
`hz4i013_g00555_5cd1f106`	50.6	54.1	54.4	52.1	53.1
`hz4i005_g00345_c7922cf1`	56.8	61.0	57.6	58.5	59.7
`hz4i002_g00785_6b2ff4aa`	59.1	62.0	58.8	61.8	59.8

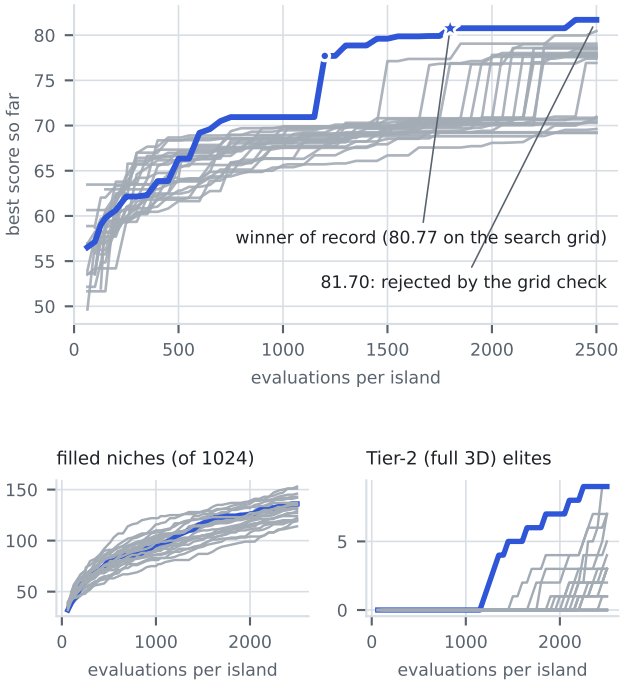


Figure 1 | Top: best measured score so far of each island (the winning island in colour; the dot marks its first full-3D best). Bottom: filled niches and Tier-2 elites per island.

74.6 g at $h = 0.15$ mm, and its Tier-1 score falls from 75.06 to 66.65 (Table 2). The two finest grids agree within 3.3 % for every design, so every number of record is measured at $h = 0.2$ mm, the grid of the print file. The final ranking ordered every elite by its record from the run, re-evaluated the top 32 at Tier 2 at $h = 0.2$ mm, and the top 8 again on the platform of record (Table 3).

6.3 The best individual

The best individual, `hz4i013_g00995_d206e3c9` (genome hash `d206e3c9f4c3`), scores 80.55 with a weighted base of 0.815 and an ergonomic multiplier of 0.988. Its grip is 64.9 mm long with a swell perimeter of 64.4 mm and a height-to-width ratio of 1.59; the guard is 11.4 mm high and the reach of the blade from the guard is 69.3 mm. As printed it weighs 59.3 g. Fig. 3 shows it; Table 4 lists its measurements.

Table 3 | Final ranking: Tier 2 at $h = 0.2$ mm on the platform of record

Rank	Design	Score	Mass (g)
1	`hz4i013_g00995_d206e3c9`	80.55	57.0
2	`hz4i002_g00995_6f4643f1`	80.44	58.7
3	`hz4i013_g00345_be45c4bc`	79.51	56.7
4	`hz4i002_g00479_76f5e0ac`	79.39	59.5
5	`hz4i013_g00555_5cd1f106`	79.27	52.1
6	`hz4i004_g00501_cb3c9e4e`	79.24	57.3
7	`hz4i002_g00905_b274d4d9`	78.94	59.6
8	`hz4i007_g00005_98a9344b`	78.77	53.7

Table 4 | Measurements of the best individual (platform of record)

Measure	Value
Mass, printed (watertight mesh)	59.3 g
Edge A per-side angle, apex / bevel	20.5–20.8/ 19.5–20.8°
Edge B per-side angle, apex / bevel	20.6–21.6/ 18.7–21.7°
Grind ends from the point, A / B	3.0 / 3.1 mm
Unsharpened plunge, A / B	9.7 / 11.9 mm
Serration depth, A / B	1.63 / 1.41 mm
Teeth, A / B	3 / 5
Weakest tooth SF, A / B	3.47 / 2.83
Solid tip SF (1.82 N·m)	0.64
Lateral capacity (plate FE)	337 N
Grip SF (formula)	0.73
Human-max twist SF (plate)	1.09
Spine 3427 N SF (plate)	1.44
Ring bearing / back capacity	8,677 / 1,686 N
Fatigue SF / yield-sequence ratio	0.94 / 0.56
Buckling factor λ	11.2

6.4 Edges and serrations

Fig. 4 plots the blade section at four stations of each edge, measured along the true edge normal with the exact decoder law. Every station of an edge lies on one line: edge A at 20.5–20.8° per side and edge B at 20.6–21.6°, with no slope steeper than 20.9° and 21.9° within 2 mm of the edge. Both edges are serrated over the rear part of the blade, and the grinds run on through the plain point region to 3.0 and 3.1 mm from the point. Under the 180 N snag and 60 N pry the weakest tooth has a safety factor of 2.83, which meets the target of 1.

7 Limits of the result

What this result is not

It is a computational design. Nothing has been printed, heat-treated, sharpened or tested, and no safety factor here is a qualification.

- **Strength against the targets.** Against a target of 1, the solid tip (0.64) is below the target; the grip under the spine load (formula 0.73, voxel FE 0.74) is below the target; the human-maximum twist (plate 1.09, voxel FE 1.69) meets the target; fatigue (0.94) is below the target; and the yield-sequence ratio (0.56 against 1.2) is below the target. Lateral capacity meets the target and the ring meets the target.
- **Mass.** The design weighs 59.3 g, which is above the 50 g goal.
- **Assumed loads.** The tooth snag and pry loads are conservative research assumptions, not measured use



Figure 2 | Best design of island island13 at each logged checkpoint: evaluation, best measured score and mass on the search grid. Same camera, light and material for every frame; geometry is the exported mesh.



Figure 3 | `hz4i013_g00995_d206e3c9`. Top: the knife, and an x-ray with every internal cavity in amber. Bottom: a section across the blade 30 mm from the point, the point, and the spine edge with its serrations. Cycles renders of the exported meshes; the blade is meshed at 0.05 mm from the exact decoder law, the rest at 0.2 mm voxels.

loads. The buckling factor is an analytic lateral-torsional estimate [7], not an eigenvalue solve.

- **Models.** Voxel finite elements resolve the printed geometry only to $h = 0.4$ mm and report percentiles, not peaks; the plate and beam surrogates of the search differ from them (Table 1). Surfaces come from marching cubes on the voxel SDF [5], except the blade, which is re-meshed at 0.05 mm from the exact decoder law.
- **Coarse search grid.** The archive was selected at $h = 0.4$ mm, which undercounts the mass of some designs (Table 2); only the final ranking is measured at $h = 0.2$ mm, so designs that the coarse grid scored low were never re-checked.
- **One shortened cohort.** The release cohort is 2,500 evaluations per island on one host; another seed or a longer run may find a better design.

8 Reproducibility

The release is commit `80e669e5b558` with decoder hash `a203278d80d2`. The winner's 179-gene row is evidence/g3/v3.0.0/winner_genome.csv; every island's logged history, the ranking records, the edge profile, the blade facts and the finite-element results sit next to it under evidence/g3/v3.0.0/, listed with their SHA-256 in SHA256SUMS. The receipt release.json is the single source of this paper and of the repository README: g3-release-tex turns it into the macros used here, and ops/ci/gen_readme.py into the README. The cohort is reproduced by building g3-evolve at that commit and running each island with its seed, --h0.4, --tier1-gate62, --tier2-gate71 and --log-every50; pilot_toolsexport rebuilds the watertight envelope and print mesh from the genome row. The binary STL files (millimetres) are listed in Table 5.

9 Conclusion

Putting the edge and serration rules into the decoder made every design the search could express a knife with a correctly ground, double-serrated two-sided blade, and the search then traded mass, strength and ergonomics within that space. The search also exposed a gap in the measurement chain: on its coarse grid, a design could look lighter than it would

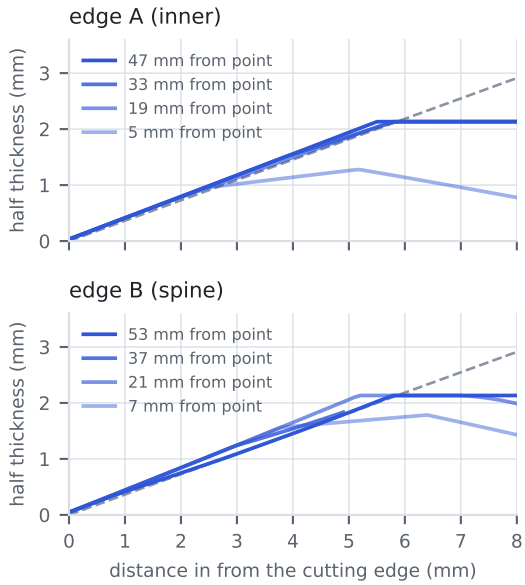


Figure 4 | Upper half of the blade section at four stations of each edge, true scale, against a 20° per-side grind (dashed). Profiles stop where a lightening slot begins.

Table 5 | CAD files of the best individual (binary STL, millimetres; each file name starts with the design identifier)

File	Triangles	MiB	Where
`AMbuild_mm.stl`	1,001,372	48	git and release
`envelope_mm.stl`	911,132	43	git and release
`AMbuild_h0.2_mm.stl`	5,373,548	256	release
`envelope_h0.2_mm.stl`	2,063,960	98	release
`envelope_fineblade_mm.stl`	3,843,106	183	release

print. Re-measuring the ranking on the print grid closed that gap for this release. The best individual of the release, `hz4i013_g00995_d206e3c9`, scores 80.55 at 59.3 g with an ergonomic multiplier of 0.988. The limits in Section 7 set the next work: a stronger sharp point, a stiffer hollow grip under the spine load, a search grid fine enough that it cannot be gamed, and a longer cohort, all measured the same way.

References

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